

**DATA 3464: Fundamentals of Data Processing**

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# Signals and images

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# Topic overview

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- Introduction to signals
- Images as 2D signals
- File formats
- Signal and image processing (for machine learning)

**Resources used:**

# What is a signal?

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*"A [continous/discrete] signal is a function of independent variables that range over [a continuum/discrete] values" - Jerry L. Prince, Medical Imaging Signals and Systems*

- Common notation:  $x(t)$  for continuous,  $x[n]$  for discrete
- Signals are **discretized** by **sampling** at some fixed interval  $dt$